THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS X — MATHEMATICS (Standard) · MID-TERM PRACTICE PAPER

Session 2026-27 · CBSE / Directorate of Education pattern · Sub code 041

Class: 10	Subject: Mathematics	Set 04	Time: 3 Hours	Max. Marks: 80

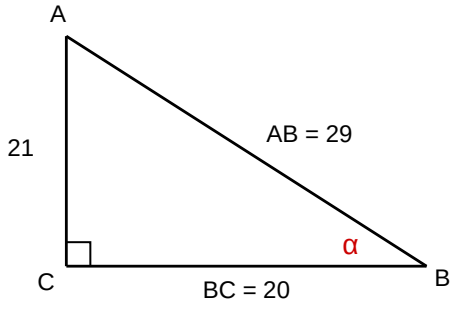
General Instructions:

1. This Question Paper has 5 Sections 'A', 'B', 'C', 'D' and 'E'.
2. Section A has 20 MCQs carrying 1 mark each (Q19–20 are Assertion–Reason questions).
3. Section B has 5 questions carrying 02 marks each.
4. Section C has 6 questions carrying 03 marks each.
5. Section D has 4 questions carrying 05 marks each.
6. Section E has 3 case based questions (04 marks each) with sub-parts of the values of 1, 1 and 2 marks respectively.
7. All questions are compulsory. However, an internal choice in 2 questions of 5 marks, 2 questions of 3 marks and 2 questions of 2 marks has been provided. An internal choice has been provided in the 2-mark questions of Section E. You have to attempt only one option out of the given options.
8. Draw neat figures wherever required. Take π = 22/7 wherever required, if not stated.
9. There is no provision of negative marking. Use of calculator is not permitted.
10. Please write down the serial number of the question before attempting it.

SECTION – A (Multiple Choice Questions · 20 × 1 = 20) — 20 questions · 20 marks

Q1–Q18 are multiple-choice questions. Q19–Q20: a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

1. From the given figure, the value of sin α is **1**



(a) $\frac{20}{29}$ (b) $\frac{21}{20}$ (c) $\frac{21}{29}$ (d) $\frac{20}{21}$

2. In ΔABC, D and E are points on AB and AC with DE ∥ BC. If AD = 2 cm, AB = 6 cm and AC = 9 cm, then AE is **1**

(a) 4 cm (b) 6 cm (c) 2 cm (d) 3 cm

3. The HCF and LCM of 6, 72 and 120 are respectively **1**

(a) 6, 360 (b) 12, 720 (c) 6, 720 (d) 12, 360

4. A natural number is chosen at random from 1 to 60. The probability that it is a multiple of 7 is **1**

(a) $\frac{7}{60}$ (b) $\frac{2}{15}$ (c) $\frac{3}{20}$ (d) $\frac{1}{7}$

5. Which of the following points CANNOT lie on the line $x + y = 0$? **1**

(a) (0, 0) (b) (2, 2) (c) (3, −3) (d) (−5, 5)

6. $(5 + \sqrt{7})(5 - \sqrt{7})$ equals **1**

(a) 32 (b) 12 (c) 18 (d) $25 - 7\sqrt{7}$

7. If the HCF of 65 and 117 is expressible as $65m - 117$, then the value of m is **1**

(a) 2 (b) 1 (c) 3 (d) 4

8. A person saves ₹ 100 in the first month and ₹ 20 more every following month. The saving in the 8th month is **1**

(a) ₹ 220 (b) ₹ 260 (c) ₹ 200 (d) ₹ 240

9. For which value of k do $3x + ky = 9$ and $6x + 8y = 18$ have infinitely many solutions? **1**

(a) 3 (b) 4 (c) 6 (d) 8

10. The value of $\sec^2\theta - \tan^2\theta$ is **1**

(a) 0 (b) −1 (c) $\sin^2\theta$ (d) 1

11. The point on the x-axis equidistant from A(−2, 0) and B(6, 0) is **1**

(a) (4, 0) (b) (0, 2) (c) (−2, 6) (d) (2, 0)

12. Two coins are tossed together. The probability of getting at most one head is **1**

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{3}{4}$ (d) 1

13. The solution of $x + y = 11$ and $x - y = 3$ is **1**

(a) x = 4, y = 7 (b) x = 8, y = 3 (c) x = 7, y = 4 (d) x = 6, y = 5

14. The LCM of two numbers is 2400 and HCF is 20. If one is 400, the other is **1**

(a) 60 (b) 120 (c) 240 (d) 100

15. A die is thrown once. The probability of getting a prime number greater than 2 is **1**

(a) $\frac{1}{2}$ (b) $\frac{1}{6}$ (c) $\frac{2}{3}$ (d) $\frac{1}{3}$

16. If $\sin\theta = \frac{3}{5}$ for an acute angle θ, then $\tan\theta$ equals **1**

(a) $\frac{4}{3}$ (b) $\frac{3}{5}$ (c) $\frac{3}{4}$ (d) $\frac{5}{4}$

17. The lines $2x + 3y = 5$ and $4x + 6y = 7$ are **1**

(a) coincident (b) parallel (c) intersecting (d) perpendicular

18. The 5th term of the AP −3, −1, 1, 3, … is **1**

(a) 5 (b) 7 (c) 4 (d) 3

19. **Assertion (A):** The point (−4, 0) lies on the x-axis. **1**
Reason (R): The y-coordinate of every point on the x-axis is zero.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

20. **Assertion (A):** The sum of two irrational numbers is always irrational. **1**
Reason (R): $\sqrt{2}$ and $-\sqrt{2}$ are irrational.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION – B (Very Short Answer Questions · 5 × 2 = 10) — 5 questions · 10 marks

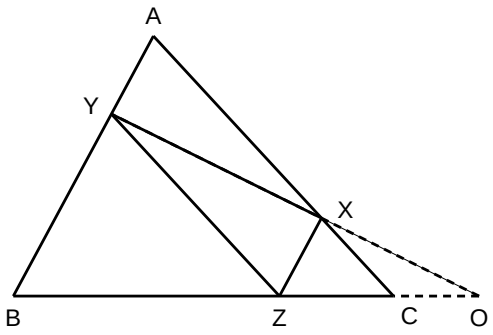
21. Show that 12^n cannot end with the digit 0 for any natural number n. **2**

OR

Two runners take 12 min and 18 min to complete a round of a park. If they start together, after how many minutes will they meet again at the start?

22. A girl 1.5 m tall casts a shadow 2 m long. At the same time a tree casts a shadow 12 m long. Find the height of the tree. **2**

23. In the figure, Z is on BC, X on AC and Y on AB. State why the triangle ratio AB/AC = BZ/ZC applies when AZ bisects ∠A, and find BZ if AB = 6, AC = 9 and BC = 10. **2**



O R

If ΔABC ~ ΔDEF, AB = 4 cm, DE = 6 cm, EF = 9 cm and FD = 12 cm, find the perimeter of ΔABC.

24. A ladder 10 m long reaches a window $5\sqrt{3}$ m above the ground. Find the angle the ladder makes with the ground. **2**

25. Find the distance between the points (2, −3) and (−4, 5). **2**

SECTION – C (Short Answer Questions · 6 × 3 = 18) — 6 questions · 18 marks

26. Prove that $3 + 2\sqrt{5}$ is irrational, given $\sqrt{5}$ is irrational. **3**

27. Solve: $3x + 4y = 10$ and $2x - 2y = 2$. **3**

28. How many terms of the AP 9, 17, 25, … must be taken so that the sum is 636? **3**

O R

The 4th term of an AP is 0. Prove that its 25th term is triple its 11th term.

29. Find the ratio in which the point P(2, y) divides A(−2, 2) and B(3, 7). Also find y. **3**

O R

Find the area of the triangle with vertices (1, −1), (−4, 6) and (−3, −5).

30. If $\cos\theta = \frac{8}{17}$, find the value of sin θ and tan θ. **3**

31. Two dice are thrown together. Find the probability that the sum of the numbers is (a) 8 (b) at least 10 (c) a perfect square. **3**

SECTION – D (Long Answer Questions · 4 × 5 = 20) — 4 questions · 20 marks

32. If a line is drawn parallel to one side of a triangle to intersect the other two sides at distinct points, prove that the other two sides are divided in the same ratio. **5**

33. The sum of the digits of a two-digit number is 9. Nine times the number is twice the number obtained by reversing the digits. Find the number. **5**

O R

Solve graphically: $x + y = 6$ and $x - y = 2$. Find the area of the triangle formed by the lines and the x-axis.

34. A man on a tower 60 m high observes two cars on the same side of the tower, in a line with its foot, at angles of depression 30° and 60°. Find the distance between the cars. (Take $\sqrt{3} = 1.732$) **5**

35. Prove that $(\sin\theta + \operatorname{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2 = 7 + \tan^2\theta + \cot^2\theta$. **5**

OR

If $\tan(A + B) = \sqrt{3}$ and $\tan(A - B) = \frac{1}{\sqrt{3}}$, $0° < A + B \leq 90°$, $A > B$, find A and B and the value of sin A cos B + cos A sin B.

SECTION – E (Case Study Based Questions · 3 × 4 = 12) — 3 questions · 12 marks

36. A school organises a mathematics fair. In a spinner game, a pointer spins on a circle divided into 20 equal sectors numbered 1 to 20.

(i) Find the probability of getting an even number. 1

(ii) Find the probability of getting a number divisible by 3. 1

(iii) Find the probability of getting a prime number greater than 10. 2

OR

(iii) Find the probability of getting a number that is neither a multiple of 4 nor of 5.

37. A village is mapped on a grid. The school S is at (2, 3), the hospital H is at (8, 11) and the market M is at (5, 7). Distances are in km.

(i) Find the distance between the school and the hospital. 1

(ii) Show that the market lies at the midpoint of SH. 1

(iii) A new point P lies on SH with SP : PH = 1 : 4. Find its coordinates. 2

OR

(iii) Find the point on the x-axis equidistant from S and H.

38. A library arranges books on a staircase-shaped rack. The bottom shelf holds 40 books and each shelf above holds 3 fewer books than the one below. There are 12 shelves.

(i) Write the first term and the common difference of the AP. 1

(ii) How many books are on the 12th shelf? 1

(iii) Find the total number of books on the rack. 2

OR

(iii) Which shelf holds 22 books?